

Accelerating Post-Quantum HQC KEM on ARMv9-A

Leveraging Vectorization via SVE2 and NEON Pipelines

PQC Crypto Research • High-Performance Systems Group

The Post-Quantum Threat

Mitigating Quantum Attacks on Classic Public-Key Cryptography via Code-Based Cryptographic Schemes

The HQC KEM Challenge

The Paradigm Shift

Unlike lattice-based schemes, Hamming Quasi-Cyclic (HQC) is a code-based key encapsulation mechanism (KEM). Its security relies on the hardness of Quasi-Cyclic Syndrome Decoding (QCSD) for binary linear codes, ensuring high resistance to quantum attacks.

Arithmetic Bottleneck

Arithmetic in HQC is defined over the binary polynomial ring. Standard fast prime-field techniques do not apply. Execution time is heavily dominated by binary polynomial multiplication, necessitating ultra-fast vectorized implementations.

HQC Parameter Sets

Parameter Set	NIST Level	Ring Dimension (n)	Hamming Weight (w)	Public Key (ek)	Ciphertext (c)
HQC-1	Level 1	17,669	66	2,241 Bytes	4,433 Bytes
HQC-3	Level 3	35,851	100	4,514 Bytes	8,978 Bytes
HQC-5	Level 5	57,637	131	7,237 Bytes	14,421 Bytes

Target Architecture: ARMv9-A

Unlocking Scalable Data-Level Parallelism with Modern SVE2 Engine Co-design

Scalable Vector Extension 2

Vector Length Agnostic (VLA)

Writing vectorized code without assuming fixed physical widths (supporting 128 to 2048 bits dynamic hardware scaling).

SVE2 & NEON Co-existence

Utilizing a balanced 128-bit SVE2 implementation operating concurrently with legacy NEON execution mode.

Predicate Registers

Using 16 predicate registers (p0-p15) to perform granular element lane-masking and dynamic hardware loops.

// SVE2 Scalable Execution Architecture

NEON Engine (Fixed Width)

Lane 3

Lane 2

Lane 1

Lane 0 (128-bit)

SVE2 Engine (Variable Scale)

Base Vector Lane Segment (128-bit)

Scalable Hardware Blocks (Up to 2048-bit Virtual Lanes)

Mathematical Engine: AFFT

√x Additive FFT (AFFT)

Uses Cantor bases instead of roots of unity to accelerate polynomials over finite fields of characteristic two $\mathbb{F}_{2^{64}}$.

⌋ Vanishing Polynomials

Possesses a highly sparse linearized algebraic structure, eliminating arbitrary modular division.

$$s_i(x) = \prod_{j \in V_i} (x - j)$$

⊞ Novel Polynomial Basis

Enables high-efficiency basis conversion and evaluation using the binary Butterfly algorithm.

$$X_i(x) = \prod_{j \geq 0} s_j(x)^{i_j}$$

Optimization Methodology



Vectorized PolyMult

Developing SVE2 instructions for fast binary polynomial multiplication, resolving computational bottlenecks in polynomial rings.



Vectorized Btfy

Structuring parallel SVE2-NEON recursive algorithms to compute Butterfly arithmetic over large Cantor bases with zero-copy states.



Secure Codecs

Highly optimized Reed-Muller and Reed-Solomon encoders and decoders, featuring constant-time sampling to resist side-channel leakage.